AGILE APPROACH – BIG DATA PROJECT

Marketplace Revenue and Delivery-Performance Pipeline | Team 6 | Checkpoint 3

GitHub repository: https://github.com/jayesh05bablr-code/Marketplace-Revenue-Delivery-Performance-Pipeline

1. Project overview

The project uses the Brazilian E-Commerce Public Dataset by Olist to design a data pipeline for marketplace revenue and delivery-performance analysis. The business objectives are to analyse monthly revenue by product category and customer state, compare delivery performance by state and seller, prepare seller scorecards, and examine the relationship between delivery timeliness and customer review scores.

The Databricks workspace is organised into six notebooks: 01_Bronze_Layer_Ingestion, 02_Silver_Layer_Transformation, 03_Data_Modeling, 04_Data_Quality_Validation, 05_Gold_Layer_Analytics and 06_Dashboard_Insights. The GitHub repository is the project's source-control and documentation location.

2. Agile working approach

The team uses an iterative, backlog-driven approach. The project is divided into small work packages aligned with the six notebook stages. Tasks are prioritised by technical dependency, assigned an owner and reviewer by the team, and assessed against clear acceptance criteria. Suggested task states are To Do, In Progress, Review and Done. The team should update its tracker as work progresses.

3. Product backlog

Work package	Activities	Acceptance evidence
01 – Bronze ingestion	Load nine Olist CSV files; preserve source values; add load timestamp and source filename; write Delta tables.	Nine Bronze tables; schema and row-count checks.
02 – Silver transformation	Cast data types; parse timestamps; standardise text; handle duplicates and missing values; translate product categories.	Silver tables with documented cleaning rules.
03 – Data modeling	Define order-item fact grain; create customer, seller, product and date dimensions; aggregate reviews at order level before joining.	Fact/dimension tables and star-schema design.
04 – Data quality	Check row counts, nulls, key uniqueness, orphan references and date consistency; record results.	Validation output and quality log.
05 – Gold analytics	Create revenue, delivery and seller-performance aggregations.	Gold analytical tables.
06 – Dashboard insights	Query Gold tables and prepare dashboard visuals and interpretations.	Dashboard-ready results and saved visualisations.

4. Iteration and sprint planning

A practical iteration sequence follows the pipeline dependencies: (1) source ingestion and profiling; (2) Silver cleaning and standardisation; (3) modeling and data-quality validation; and (4) Gold analytics and dashboard preparation. This is a suggested work breakdown, not a claim about the team's actual sprint dates or completed sprint ceremonies. Record the real sprint duration, dates, selected backlog items and outcomes in the team tracker.

For each iteration, agree on a sprint goal, break work into tasks, identify dependencies, assign an owner and reviewer, and define acceptance criteria. During review, demonstrate working outputs such as Delta tables and row counts, cleaned schemas, fact/dimension tables, validation results and analytical aggregates.

5. Team structure and responsibilities

The approved project plan lists five team members: Jayesh Santosh Shinde, Yash Parmar, Kirti Jain, Jigyas Niraj Borah and Prishita Girdhar. Individual notebook ownership is not confirmed by the available project information. The team should record actual task owners and reviewers rather than infer assignments. Suggested responsibility areas include backlog coordination, data engineering, modeling and quality, analytics/dashboard development, and documentation/code review.

6. Collaboration and version control

The project repository is: https://github.com/jayesh05bablr-code/Marketplace-Revenue-Delivery-Performance-Pipeline. Use it to maintain the README, exported Databricks notebooks, documentation and architecture artifacts. Commit changes with descriptive messages, review code before submission, and keep the repository aligned with the workspace version. Do not

commit passwords, access tokens or cloud credentials.

7. Risks and mitigation

Data quality: profile nulls, duplicates and invalid values before deciding how to handle them; legitimate missing delivery timestamps should remain distinguishable from data errors. Schema/type issues: validate casts before joins. Join fan-out: preserve the order-item grain and aggregate order-level reviews before joining. Environment differences: document catalog, schema and volume settings. Time constraints: prioritise the required transformation code and Agile write-up for Checkpoint 3; describe later stages as planned until executed and verified.

8. Review and retrospective

At the end of each iteration, discuss what went well, what caused delays, which assumptions were incorrect and what should change. Record specific improvement actions and owners. Potential improvements include reusable read/write functions, schema checks before ingestion, consistent naming conventions, automated quality logging and peer review of notebook outputs.

9. Repository and implementation status

The supplied GitHub repository URL was checked for this update. At the time of review, its main branch displayed a README.md containing only the repository heading; the repository page did not show the six notebook files or documentation PDFs. The six notebook names are confirmed by the Databricks Workspace screenshot, not by the current GitHub file listing. Push the exported notebooks and documentation to GitHub before describing them as repository contents.

This document describes the Agile approach for the six-notebook project. Update it with actual sprint dates, task assignments, completed work, review outcomes and retrospective actions before final submission.

Dataset note: The project plan identifies the Olist dataset licence as CC BY-NC-SA 4.0. Preserve attribution and verify the dataset page's licence conditions before redistributing raw or derived data.